
RDTII Classification Tool – User Guide

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Automated_Legal_Document_Analysis_for_RDTII_Indicators
March 16, 2026

Introduction

This guide explains how to use the RDTII Classification Tool to analyze legal documents. The system is designed to help lawyers, legal analysts, and policy researchers process legal texts, identify relevant RDTII indicators, and generate structured results for review. The guide provides step-by-step instructions to run the system and interpret the output in a practical and efficient way. It is designed to be used without technical or programming knowledge.

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I Purpose of the System

The RDTII Classification Tool is designed to support the analysis of legal documents in a simple and efficient way.

- The tool reads legal documents in PDF format
- It splits the documents into structured sections
- It predicts relevant RDTII indicators for each section
- It exports the results into a table for review

The system helps users quickly identify relevant regulatory content and supports further legal and policy analysis.

2 Installing Python and PyCharm

This section explains how to download and install Python and PyCharm before running the RDTII Classification Tool.

2.1 Downloading and Installing Python

1. Open the official Python website in your browser:
<https://www.python.org/downloads/>
2. On the download page, click the yellow **Download** button to download the latest version of Python for your operating system (Figure 1).



Figure 1: Python download page

3. If you need to choose a specific version, scroll down to view available Python releases. Select a stable version (Python 3.9 or above is recommended) and click **Download** (Figure 2).

Looking for a specific release?

Python releases by version number:

Release version	Release date		Click for more
Python 3.14.3	Feb. 3, 2026	Download	Release notes
Python 3.14.2	Dec. 5, 2025	Download	Release notes
Python 3.14.1	Dec. 2, 2025	Download	Release notes
Python 3.14.0	Oct. 7, 2025	Download	Release notes
Python 3.13.12	Feb. 3, 2026	Download	Release notes
Python 3.13.11	Dec. 5, 2025	Download	Release notes
Python 3.13.10	Dec. 2, 2025	Download	Release notes
Python 3.13.9	Oct. 14, 2025	Download	Release notes

[View older releases](#)

Figure 2: Selecting a specific Python version

4. Scroll further to the **Files** section and select the correct installer for your operating system (Figure 3):
 - macOS: Download macOS installer
 - Windows: Download Python install manager

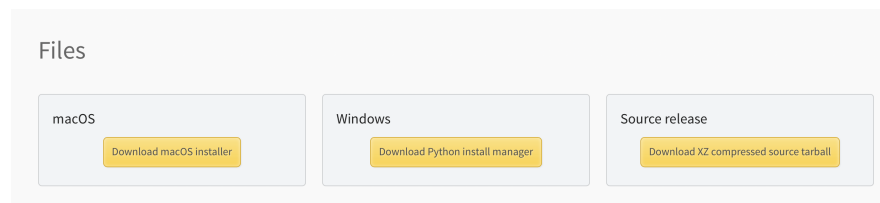


Figure 3: Selecting the correct installer file

5. After the installer is downloaded, open the Python installer. The welcome screen will appear. Click **Continue** to proceed (Figure 4).

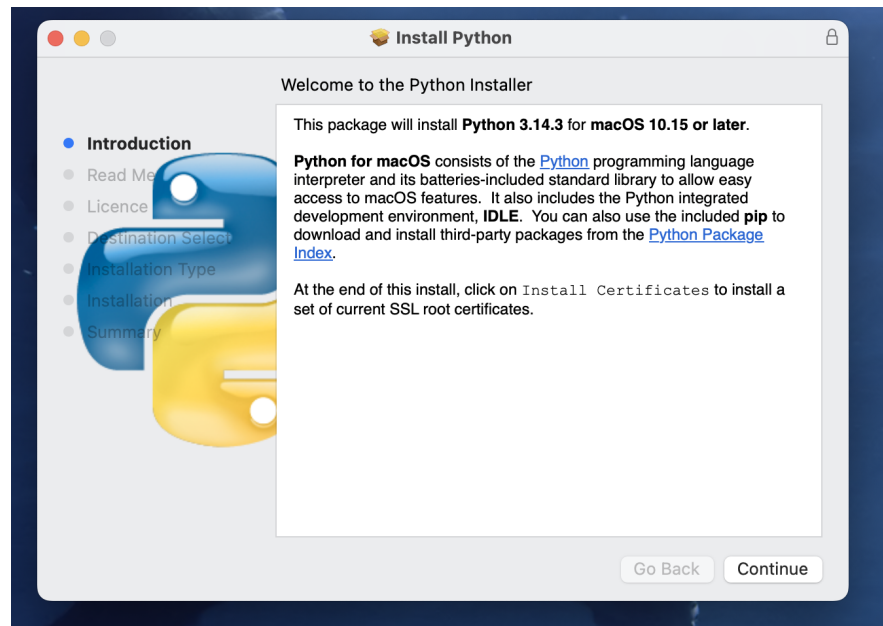


Figure 4: Python installer welcome screen

6. Read through the licence screen if needed. When the licence agreement pop-up appears, click **Agree** to continue the installation (Figure 5).

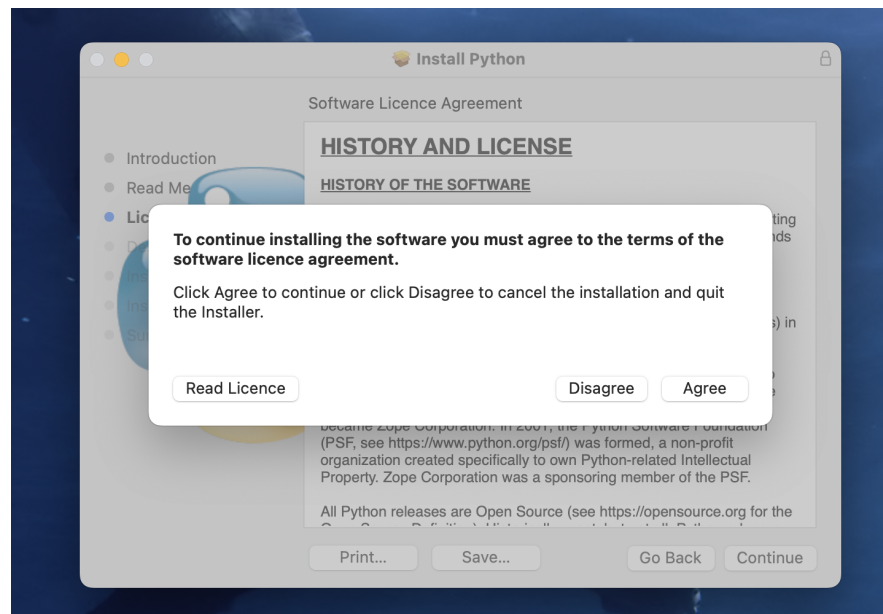


Figure 5: Accepting the Python licence agreement

7. Continue the installation using the default settings until the installation is completed successfully.
8. When the installation is finished, the installer will show a completion message. Click **Close** to exit the installer.
9. A message may appear asking whether to move the installer to the Bin. Click either **Keep** or **Move to Bin**. This does not affect the Python installation (Figure 6).

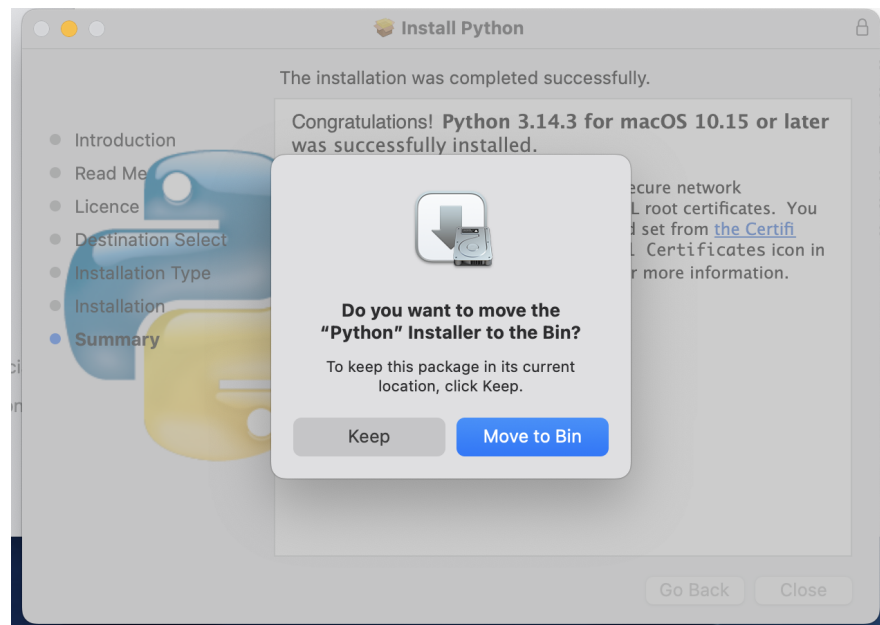


Figure 6: Optional message after Python installation

2.2 Downloading and Installing PyCharm

1. Open the official PyCharm website in your browser:
<https://www.jetbrains.com/pycharm/download/>
2. On the download page, select the correct operating system if needed, then click **Download** (Figure 7).
3. If the download options appear, choose the correct installer version for your computer, such as .dmg (Apple Silicon) or .dmg (Intel) on macOS (Figure 7).

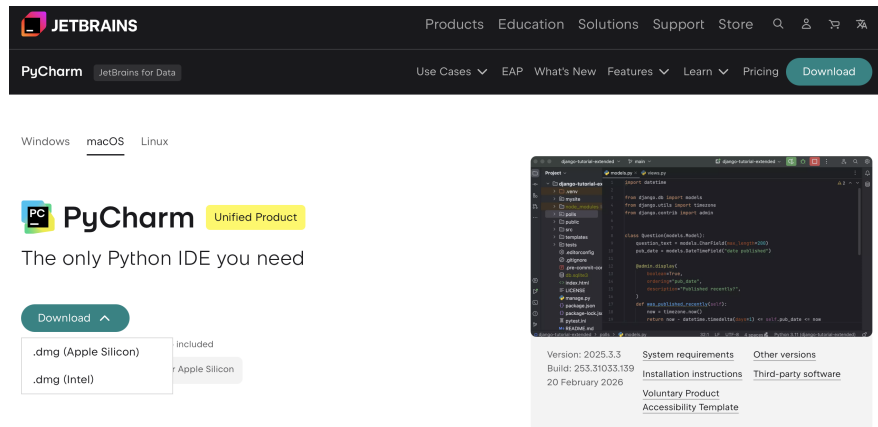


Figure 7: PyCharm download page

4. After the installer is downloaded, open the .dmg file. A window will appear showing the PyCharm icon and the Applications folder. Drag the PyCharm icon into the **Applications** folder (Figure 8).

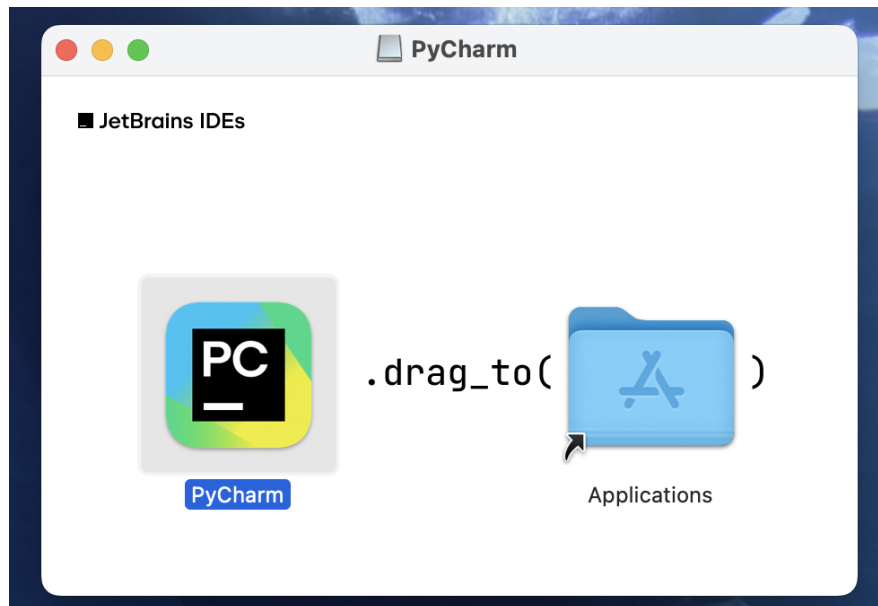


Figure 8: Installing PyCharm by dragging it into Applications

5. Open **Applications** and double-click the **PyCharm** icon to launch the program (Figure 9).



PyCharm

Figure 9: PyCharm application icon

Python must be installed before running the project. PyCharm is the interface used to open the project and run the scripts more easily.

3 Setting Up the Project Folder

This section explains how to download the project folder, extract it, and open it in PyCharm.

3.1 Downloading the Project Folder

1. Open the shared project folder (e.g. OneDrive or SharePoint).
2. Locate the project folder (e.g. BERT_training).
3. Right-click on the folder and select **Download** (Figure 10).

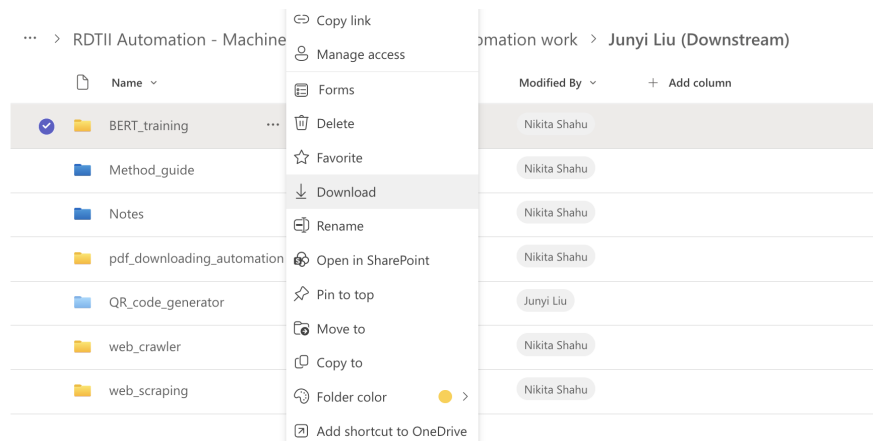


Figure 10: Downloading the project folder

4. The folder will be downloaded as a compressed file (ZIP format), e.g. BERT_training.zip.

3.2 Extracting the Project Folder

1. Locate the downloaded ZIP file in your **Downloads** folder (Figure 11).

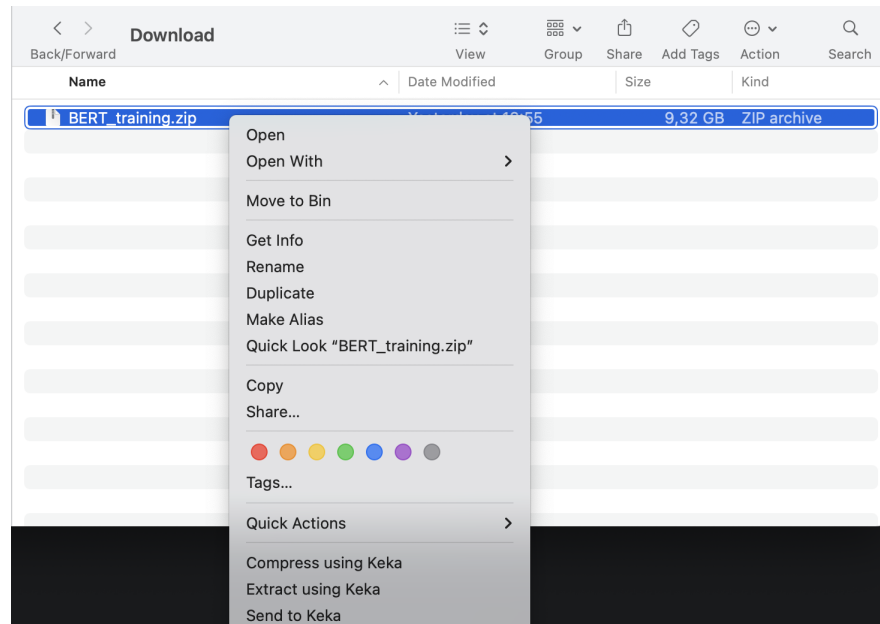


Figure 11: Downloaded ZIP file

2. Right-click on the file and choose an extraction option. See Section 3.4 for recommended tools.
3. After extraction, a new folder (e.g. BERT_training) will appear

3.3 Opening the Project in PyCharm

1. Launch **PyCharm**.
2. On the welcome screen, click **Open**.

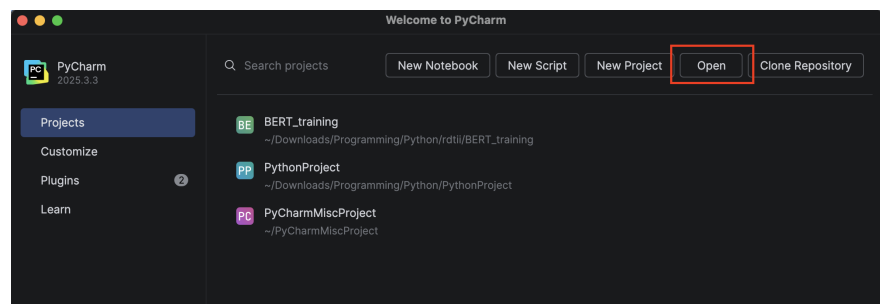


Figure 12: PyCharm welcome screen

3. Navigate to the extracted folder (e.g. BERT_training) and click **Open**.

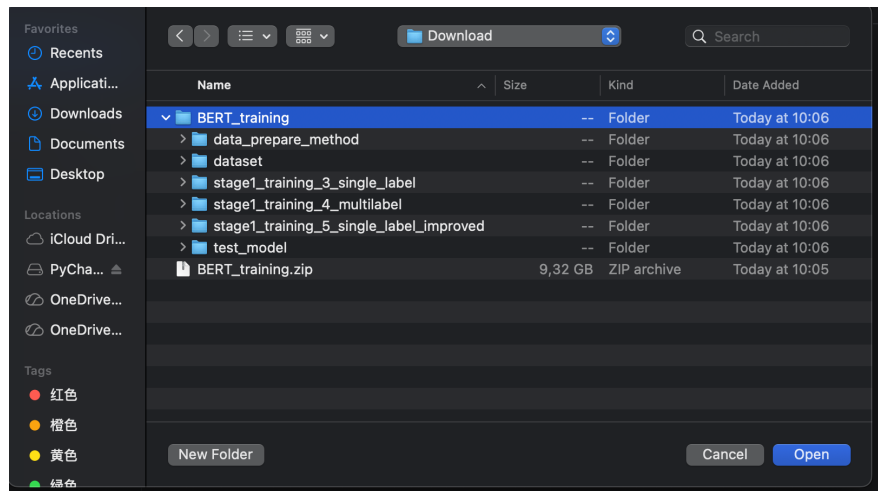


Figure 13: Selecting the project folder

4. A security prompt will appear. Click **Trust Project** to continue (Figure 14).

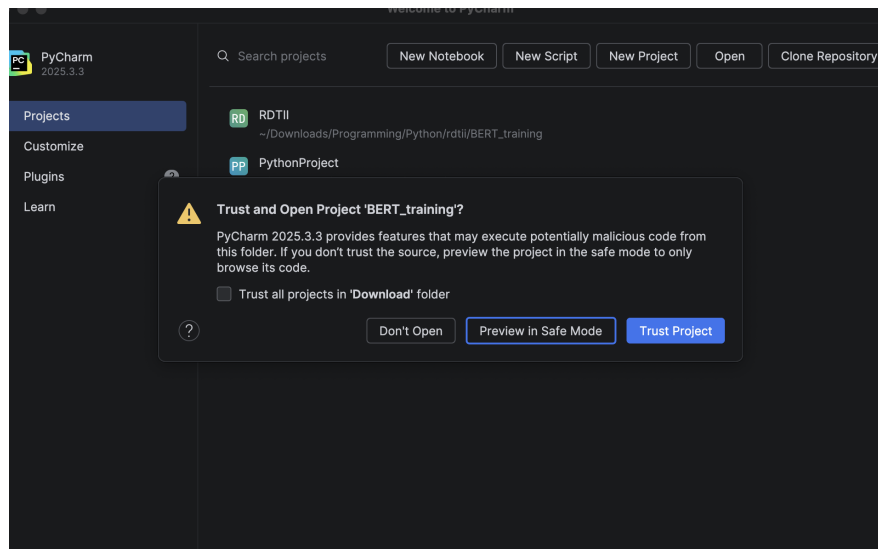


Figure 14: Trust project prompt

5. The project will open in PyCharm, and the folder structure will be displayed (Figure 15).

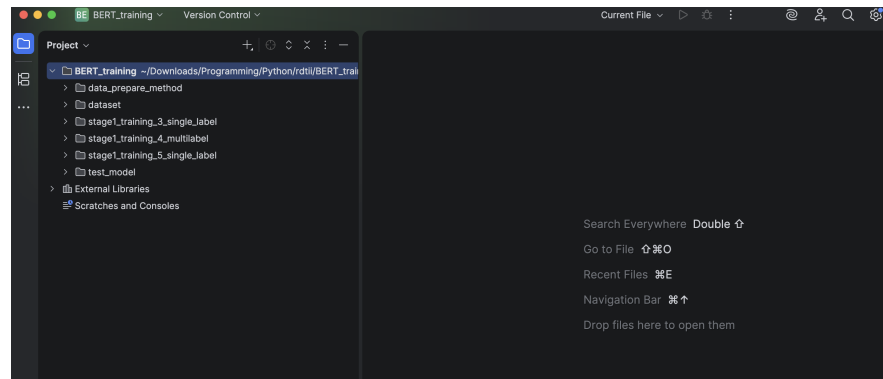


Figure 15: Project opened in PyCharm

3.4 Recommended Extraction Tools

- macOS: Keka Download: <https://www.keka.io/en/>



Figure 16: Keka file archiver (macOS)

- Windows: 7-Zip Download: <https://www.7-zip.org>

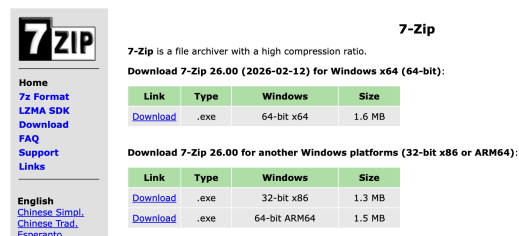


Figure 17: 7-Zip download page (Windows)

The project must be **extracted before opening in PyCharm**. Do not open the .zip file directly.

4 Understanding the System

The system processes legal documents through a step-by-step pipeline, from raw PDF input to structured classification output.

4.1 Quick Guide to Workflow

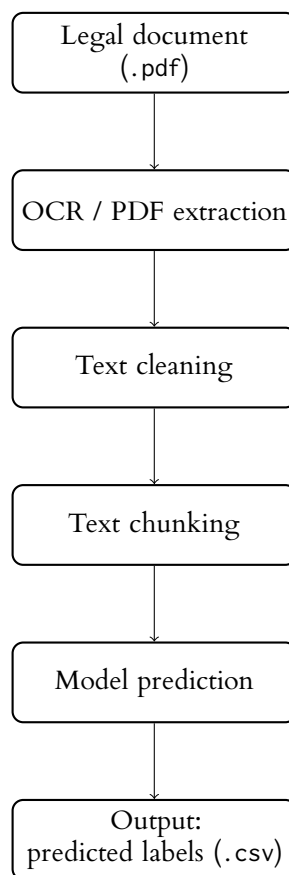


Figure 18: System workflow for legal document processing

4.2 Relevant Scripts and Folder Structure

For practical use, users only need to understand the following folders and scripts.

```
BERT_training
|-- data_prepare_method
|   |-- ocr.py
|   |-- pdf_reader.py
|   |-- clean_text_json.py
|   |-- chunking_by_structure.py
|-- dataset
|   |-- pre_process_new_data
|       |-- raw      (input PDF files)
|       |-- interim  (extracted text)
|       |-- cleaned  (cleaned text)
|       |-- chunked  (structured text chunks)
|-- test_model
|   |-- test_output  (final results)
|   |-- model3_smartchunked_fullldoc.py
|   |-- model4_multilabel_smartchunked_fullldoc.py
|   |-- model5_smartchunked_fullldoc.py
```

Additional Explanation (for Reference)

The following parts of the system are shown for completeness but are **not** required for normal use.

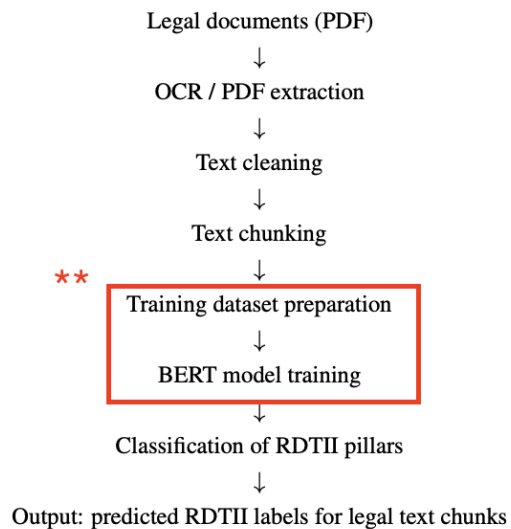


Figure 19: Training stage in the full workflow (highlighted)

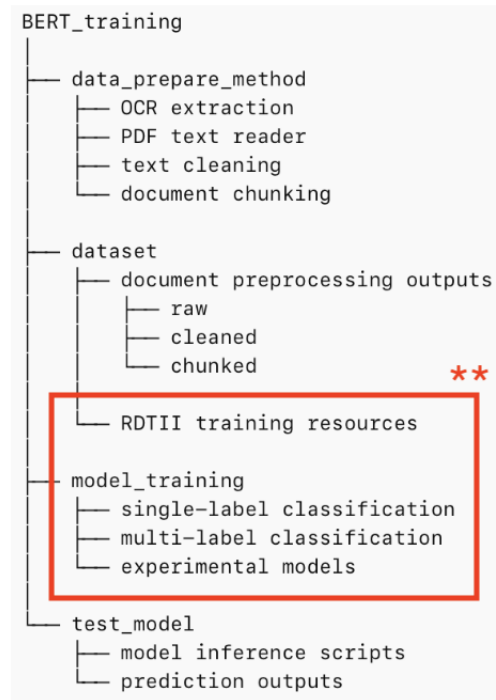


Figure 20: Training-related folders in the repository (highlighted)

Explanation:

- The highlighted part in the workflow (*Training dataset preparation* and *BERT model training*) represents the **model development stage**.
- During this stage, labeled datasets are prepared and used to train classification models.
- In the repository, this corresponds to:
 - RDIII_training_ds: contains training datasets and resources
 - model_training: contains trained models and training configurations
- These steps have already been completed in advance.
- Users do **not** need to run any training or modify these folders.
- For normal usage, users only need to run the scripts in test_model.

5 Text Extraction

This step prepares the system to read PDF documents and perform OCR (text extraction).

5.1 OCR and PDF Setup

If a package is missing, check the error message

If you run the OCR script before installing the required packages, you may see an error such as:

```
ModuleNotFoundError: No module named 'fitz'
```

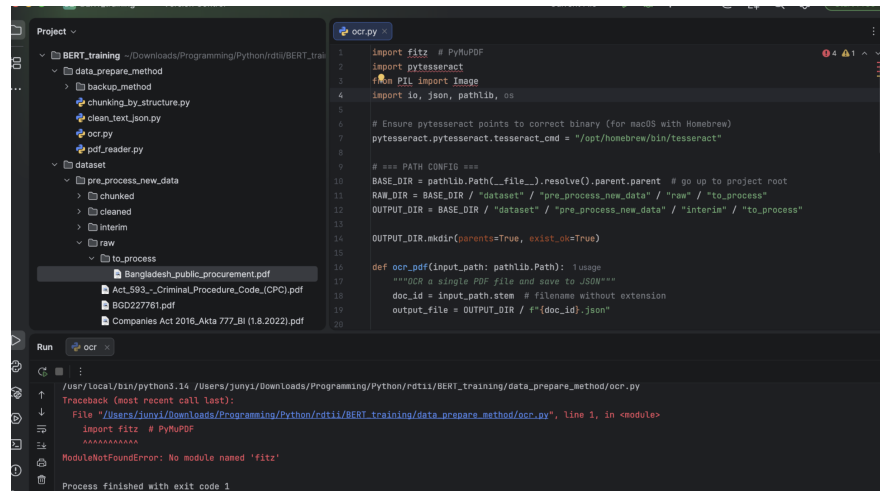


Figure 21: Error shown when PyMuPDF is not installed

This happens because the module `fitz` is provided by the PyMuPDF package. You should install PyMuPDF, not `fitz` directly.

Step 1: Open the Terminal in PyCharm

- In PyCharm, open the **Terminal** tab at the bottom of the window.
- You will see a command line where you can type installation commands.

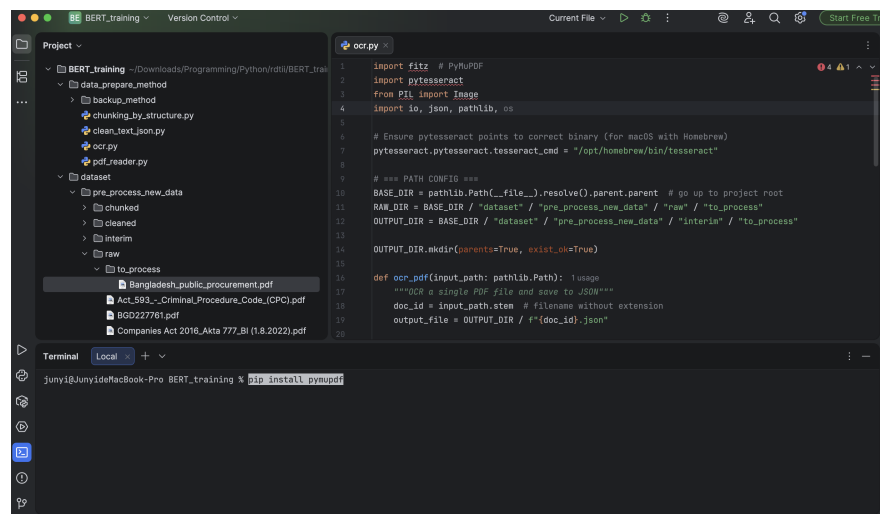
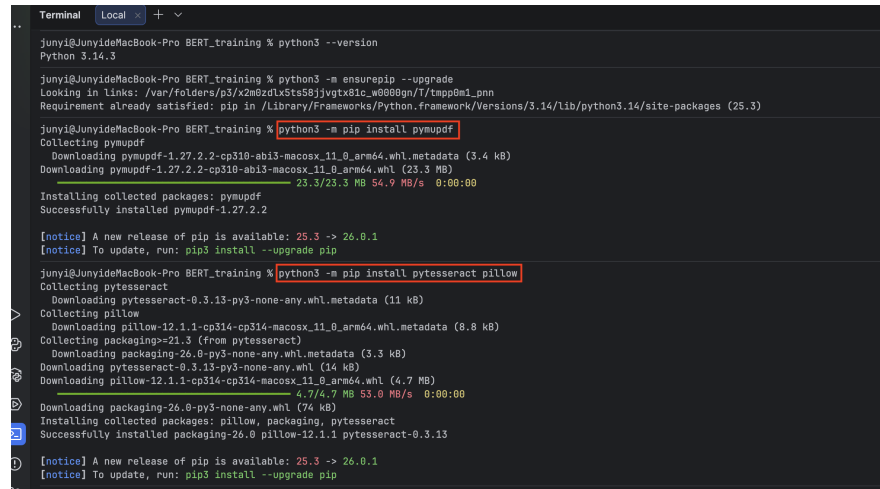


Figure 22: PyCharm terminal where installation commands are entered

Step 2: Install the required packages

Run the following commands in the terminal:

```
python3 -m pip install pymupdf
python3 -m pip install pytesseract pillow
```

A terminal window showing the installation of three Python packages. The user runs 'python3 --version' which returns 'Python 3.14.3'. Then they run 'python3 -m pip install pymupdf', which shows progress bars for downloading and installing pymupdf-1.27.2.2. Next, they run 'python3 -m pip install pytesseract pillow', which shows progress bars for downloading and installing pytesseract-0.3.13, pillow-12.1.1, and packaging-26.0. The terminal output includes version information, download progress, and successful installation messages for each package. There are also notices about newer versions of pip available.

```
Terminal Local +
junyi@JunyideMacBook-Pro BERT_training % python3 --version
Python 3.14.3

junyi@JunyideMacBook-Pro BERT_training % python3 -m pip install --upgrade
Looking in links: /var/folders/p3/x2m8zdLxStsS8jyvgtx81c_w888gn/1/tmp8m1_pnn
Requirement already satisfied: pip in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (25.3)

junyi@JunyideMacBook-Pro BERT_training % python3 -m pip install pymupdf
Collecting pymupdf
  Downloading pymupdf-1.27.2.2-cp310-abi3-macosx_11_0_arm64.whl.metadata (3.4 kB)
  Downloading pymupdf-1.27.2.2-cp310-abi3-macosx_11_0_arm64.whl (23.3 MB)
    23.3/23.3 MB 54.9 MB/s 0:00:00
Installing collected packages: pymupdf
Successfully installed pymupdf-1.27.2.2

[notice] A new release of pip is available: 25.3 -> 26.0.1
[notice] To update, run: pip3 install --upgrade pip

junyi@JunyideMacBook-Pro BERT_training % python3 -m pip install pytesseract pillow
Collecting pytesseract
  Downloading pytesseract-0.3.13-py3-none-any.whl.metadata (11 kB)
Collecting pillow
  Downloading pillow-12.1.1-cp314-cp314-macosx_11_0_arm64.whl.metadata (8.8 kB)
Collecting packaging>=21.3 (from pytesseract)
  Downloading packaging-26.0-py3-none-any.whl.metadata (3.3 kB)
  Downloading packaging-26.0-py3-none-any.whl (14 kB)
  Downloading pillow-12.1.1-cp314-cp314-macosx_11_0_arm64.whl (4.7 MB)
    4.7/4.7 MB 53.0 MB/s 0:00:00
  Downloading packaging-26.0-py3-none-any.whl (74 kB)
Installing collected packages: pillow, packaging, pytesseract
Successfully installed packaging-26.0 pillow-12.1.1 pytesseract-0.3.13

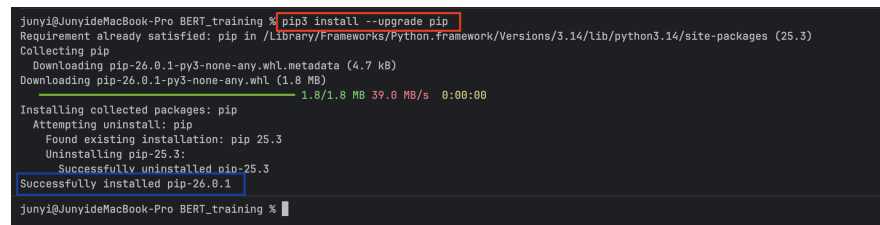
[notice] A new release of pip is available: 25.3 -> 26.0.1
[notice] To update, run: pip3 install --upgrade pip
```

Figure 23: Installing PyMuPDF, pytesseract, and pillow in the terminal

Step 3: Upgrade pip (recommended)

To avoid installation problems, it is recommended to upgrade pip:

```
pip3 install --upgrade pip
```

A terminal window showing the upgrade of pip. The user runs 'pip3 install --upgrade pip'. The terminal output shows that pip-25.3 is being uninstalled and pip-26.0.1 is being installed. Progress bars are shown for the download and installation of pip-26.0.1.

```
junyi@JunyideMacBook-Pro BERT_training % pip3 install --upgrade pip
Requirement already satisfied: pip in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (25.3)
Collecting pip
  Downloading pip-26.0.1-py3-none-any.whl.metadata (4.7 kB)
  Downloading pip-26.0.1-py3-none-any.whl (1.8 MB)
    1.8/1.8 MB 39.0 MB/s 0:00:00
Installing collected packages: pip
  Attempting to uninstall: pip
    Found existing installation: pip 25.3
    Uninstalling pip-25.3:
      Successfully uninstalled pip-25.3
  Successfully installed pip-26.0.1

junyi@JunyideMacBook-Pro BERT_training %
```

Figure 24: Upgrading pip in the terminal

Explanation of packages

- pymupdf (fitz): reads and extracts content from PDF files
- pytesseract: performs OCR (text recognition)
- pillow: supports image processing for OCR

This setup only needs to be completed once. After installation, the OCR and PDF reader scripts can run normally.

5.2 Running OCR and PDF Extraction

This section explains how to place the raw legal PDF files into the correct folder, run the OCR and PDF extraction scripts in PyCharm, and check the generated output files.

Before running the system, it is important to place the legal documents in the correct input folders and understand how the preprocessing pipeline works.

1. Place documents in the correct input folders

Go to the folder:

```
dataset/pre_process_new_data/raw/to_process
```

Inside this folder, there are two subfolders:

- digital_pdfs — for normal PDF files with selectable text
- scanned_docs — for scanned or image-based documents

Drag and drop your files into the appropriate folder (Figure 25).

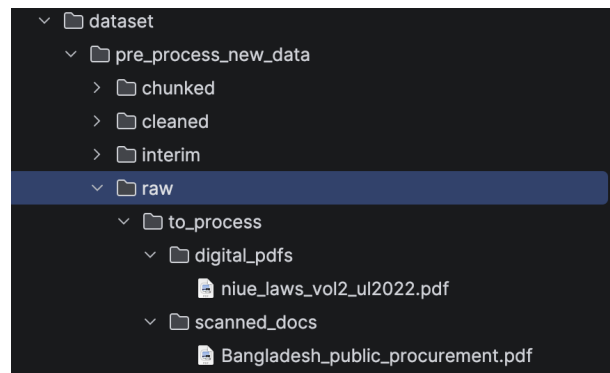


Figure 25: Input folder for raw legal PDF files

2. Open the OCR script

In the project panel, open the folder data_prepare_method.

Double-click the file ocr.py to open the OCR script (Figure 26).

The OCR script is used for scanned PDF files or image-based legal documents that cannot be read directly as machine text.

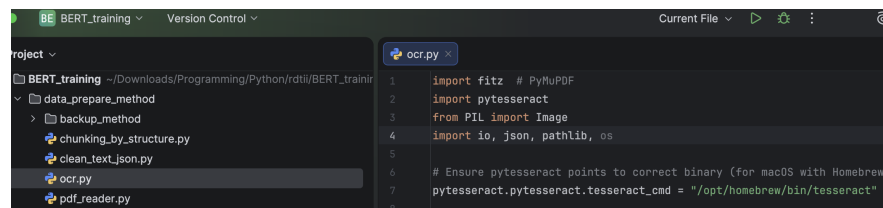


Figure 26: Opening the OCR script in PyCharm

3. Run the OCR script

After opening `ocr.py`, make sure the script tab is active.

At the top of the PyCharm window, change the run option to **Current File** if needed.

Click the green **Run** button to execute the script.

The system will scan the files placed in the raw input folder and process them using OCR.

When the script finishes successfully, the run panel will display a completion message similar to the one shown in Figure 27.

```

/usr/local/bin/python3.14 /Users/junyi/Downloads/Programming/Python/rdt11/BERT_training/data_prepare_method/ocr.py
Found 1 PDF(s) in /Users/junyi/Downloads/Programming/Python/rdt11/BERT_training/dataset/pre_process_new_data/raw/to_process
OCR complete -> /Users/junyi/Downloads/Programming/Python/rdt11/BERT_training/dataset/pre_process_new_data/interim/to_process/Bangladesh_public_procurement.json (3)
Process finished with exit code 0

```

Figure 27: OCR script completed successfully

4. Run the PDF reader script (for digital files)

In the same folder `data_prepare_method`, double-click the file `pdf_reader.py` to open it.

The PDF reader script is used for digital PDF files that already contain machine-readable text.

Make sure the `pdf_reader.py` tab is active, choose **Current File** at the top if needed, and click the green **Run** button (Figure 28).

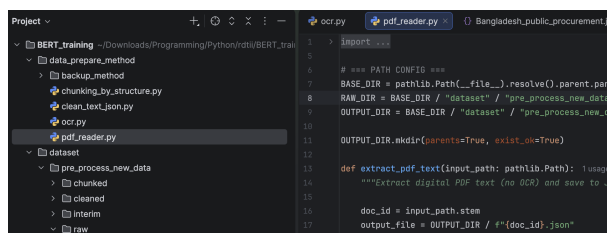


Figure 28: Opening and running the PDF reader script

5. Locate the output directory

After the extraction is completed, the generated output files will be saved in the folder:

`dataset/pre_process_new_data/interim/to_process`

In the project panel, expand the folder `interim` and then `to_process`.

The extracted files will appear as `.json` files in this folder (Figure 29).

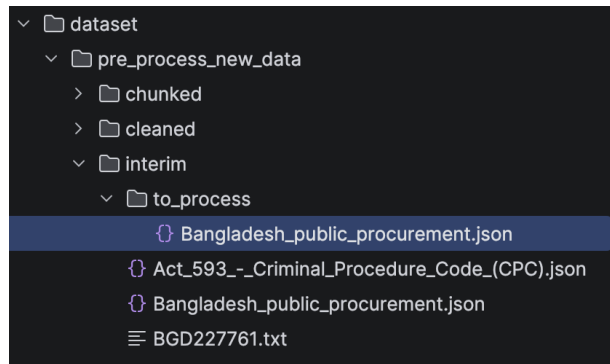


Figure 29: Output folder containing extracted interim files

6. View the extracted output

Double-click any generated .json file in the output folder to open it in the editor.

Each output file contains the extracted text page by page in JSON format.

7. Compare with the original legal document

To check whether the extraction result is correct, compare the output JSON file with the original legal PDF.

Figure 30 shows the original Bangladesh legal document page.

Figure 31 shows the corresponding extracted JSON output.

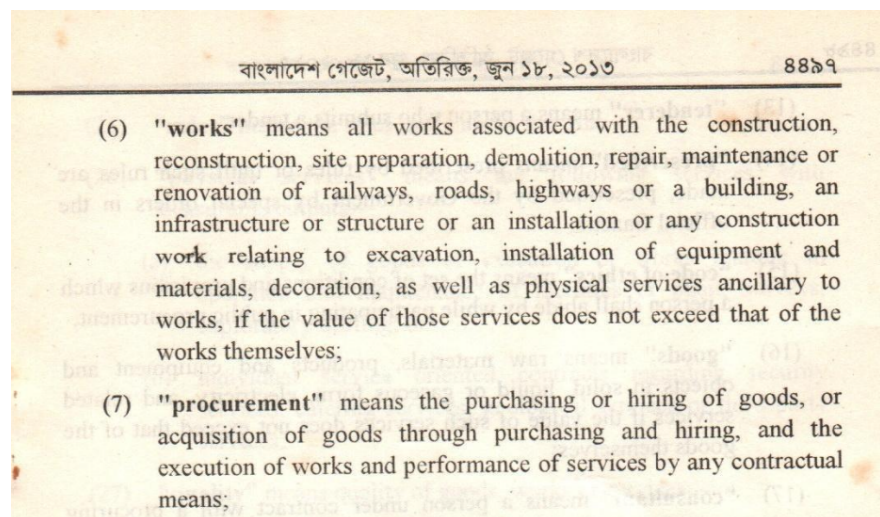


Figure 30: Original Bangladesh legal document

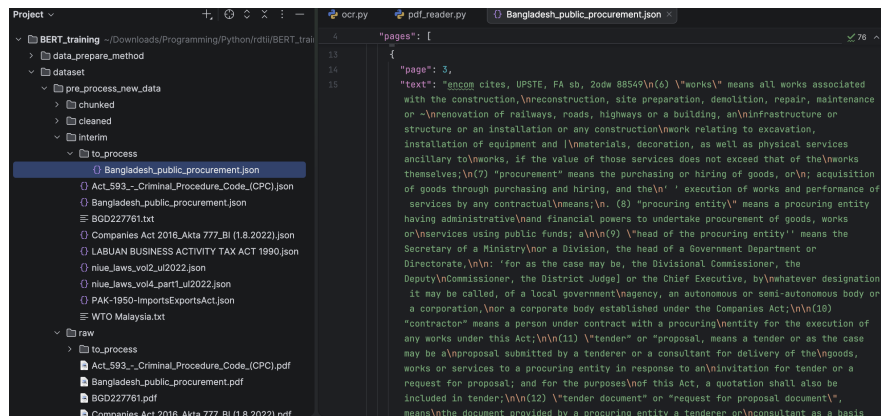


Figure 31: Extracted Bangladesh interim output

Figure 32 and Figure 33 show another example using the Niue legal document.

- Niue Legislation 2019
- 3 Act to bind the Crown**
This Act shall bind the Crown.
- PART 1**
ACTIONS FOR DAMAGES
- 4 Right of action when death is caused by negligence**
(1) Where the death of a person is caused by any wrongful act, neglect, or default, and the act, neglect, or default is such as would (if death had not ensued) have entitled the party injured to maintain an action and recover damages in respect of it, the person who would have been liable if death had not ensued shall be liable to an action for damages, notwithstanding the death of the person injured, and although the death was caused under such circumstances as to amount in law to a crime.
(2) [Spent]
(3) Not more than one action shall lie under this Act for the same subject-matter of complaint.
- 5 Action to be for benefit of family**
Every such action shall be for the benefit of the wife or husband and the parents and children of the person whose death has been so caused.

Figure 32: Original Niue legal document


```

"page": 8,
"text": "Niue Legislation 2019 \n2 \n3 \nAct to bind the Crown \nThis Act shall bind the
Crown. \nPART 1 \nACTIONS FOR DAMAGES \n4 \nRight of action when death is caused by
negligence \n(1) \nWhere the death of a person is caused by any wrongful act, neglect, or
\ndefault, and the act, neglect, or default is such as would (if death had not ensued)
have \nentitled the party injured to maintain an action and recover damages in respect of
it, the \nperson who would have been liable if death had not ensued shall be liable to
an action \nfor damages, notwithstanding the death of the person injured, and although
the death \nwas caused under such circumstances as to amount in law to a crime. \n(2)
\n[Spent] \n(3) \nNot more than one action shall lie under this Act for the same
subject-\nmatter of complaint. \n5 \nAction to be for benefit of family \nEvery such
action shall be for the benefit of the wife or husband and the parents and \nchildren of
the person whose death has been so caused. \n6 \nPersons who may bring action \n(1) \n(a)
Every such action shall be brought by and in the name of the executor \nor administrato

```

Figure 33: Extracted Niue interim output

8. Adjust the viewer if needed

If the JSON output is difficult to read, the PyCharm viewer can be adjusted.

In the top menu bar, select:

View > Active Editor

From this menu, you may adjust the display options, such as enabling soft wrap, showing line numbers, or changing the font size (Figure 34).

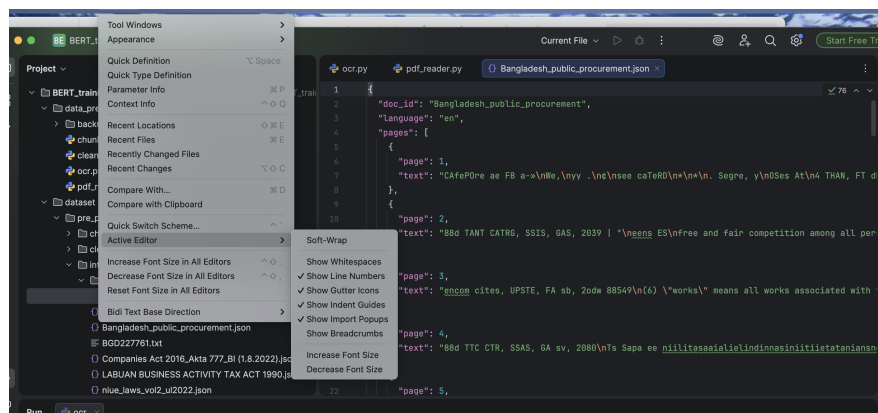


Figure 34: Adjusting the viewer settings in PyCharm

Use ocr.py for scanned or image-based PDF files. Use pdf_reader.py for digital PDF files that already contain selectable text. Always check the interim JSON output before proceeding to the next stage.

6 Text cleaning and chunking

This section explains how to process the extracted data through cleaning and chunking.

1. Prepare the input data

Make sure the extracted files are already located in:

```
dataset/pre_process_new_data/interim/to_process
```

These files will be used as input for the cleaning step.

All files in this folder will be processed automatically. You do not need to select files one by one.

2. Understand the processing pipeline

The data will move through the following folders:

- interim/to_process → extracted data
- cleaned/to_process → cleaned data
- chunked/to_process → final structured output

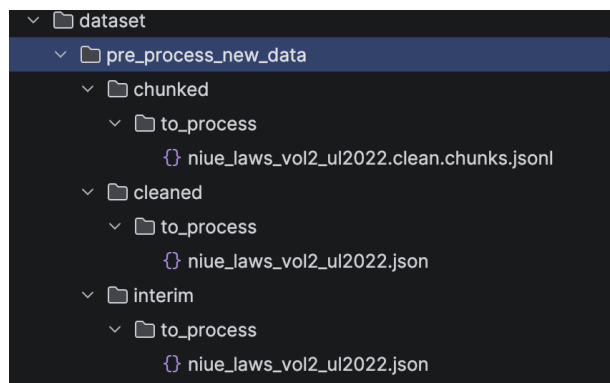


Figure 35: Processing folders: interim, cleaned, and chunked

Each step uses the output of the previous one.

3. Run the cleaning script

Open the folder:

```
data_prepare_method
```

Double-click:

```
clean_text_json.py
```

Click the green **Run** button.

The script will process all files in:

```
interim/to_process
```

and save the cleaned results in:

cleaned/to_process

4. Run the chunking script

Next, open:

chunking_by_structure.py

Click the green  button again.

The script will process all cleaned files and save results in:

chunked/to_process

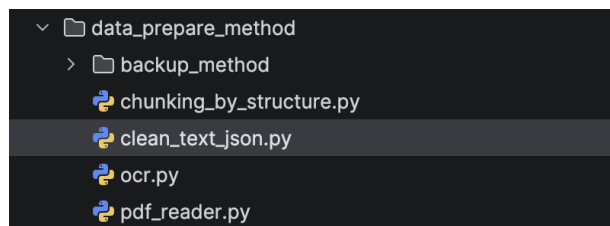


Figure 36: Cleaning and chunking scripts in the project folder

5. Check the final output

The final output files are in .jsonl format and contain structured text segments ready for model prediction.

Run the scripts in order: cleaning first, then chunking. Each script only needs to be run once and will process all files automatically.

7 Running the Model

This section explains how to run the trained model on the processed data and generate prediction results. The model takes the chunked text as input and produces structured outputs, including predicted labels and confidence scores. The following steps guide users through selecting a model, running the script, and viewing the results.

7.1 Installing Required Packages

If an error occurs when running the model script (e.g., `ModuleNotFoundError: No module named 'torch'` as shown in Figure 37), it means that the required Python packages are not installed.

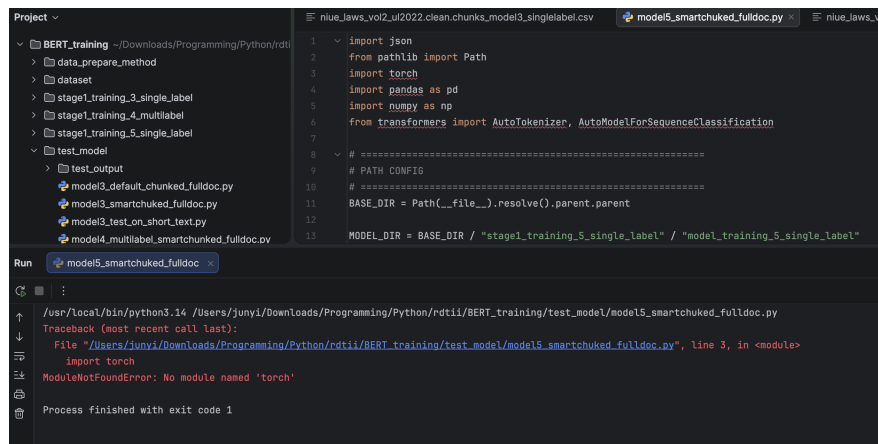


Figure 37: Example of missing package error when running the model

1. Install required packages

Similar to Section 5.1, open the terminal and run:

```
pip install torch
pip install pandas numpy transformers
```

2. If installation is not successful

Try using the full Python command:

```
python3.14 -m pip install torch
python3.14 -m pip install pandas numpy transformers
```

Make sure the packages are installed in the same Python environment used by PyCharm. Otherwise, the script may still not recognize them.

7.2 Selecting a Model Script

The folder `test_model` contains several model scripts for different prediction tasks.

1. Locate the model scripts

Open the folder:

`test_model`

This folder contains different model scripts and the output folder (Figure 38).

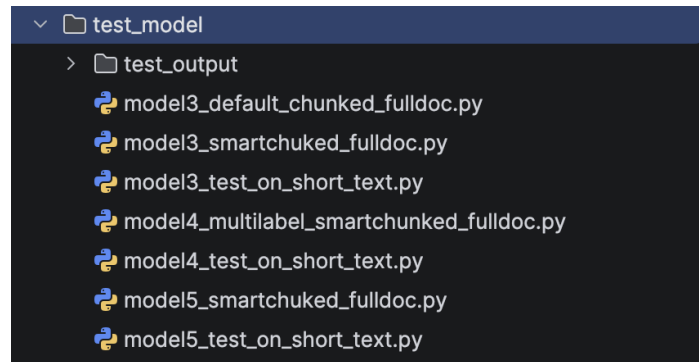


Figure 38: Model scripts available in the test_model folder

2. Main scripts for standard document prediction

For most normal use cases, the following three scripts are the main options:

- model3_smartchunked_fulldoc.py
- model4_multilabel_smartchunked_fulldoc.py
- model5_smartchunked_fulldoc.py

These scripts are designed for documents that have already been processed through the smart chunking pipeline.

3. Difference between Model 3, Model 4, and Model 5

- Model 4 is a multi-label model. It can assign more than one label to the same text segment.
- Model 3 and Model 5 are single-label models. They assign one main label to each text segment.

At the current stage, **Model 5** may be recommended for standard single-label prediction. However, Model 3 and Model 5 are still relatively similar in function, so further diagnostic comparison may still be needed.

4. Scripts for other special cases

Some scripts are designed for specific situations rather than standard full-document prediction:

- model3_default_chunked_fulldoc.py is used for default chunked input rather than smart chunked input.
- Scripts ending in test_on_short_text.py are used for short text prediction.

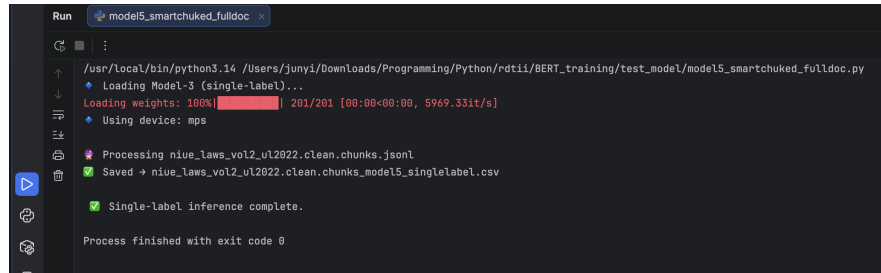
These scripts can be useful in special cases, for example:

- when the chunking result is not ideal
- when a user wants to test one paragraph quickly
- when a short legal passage needs to be checked separately

For normal document analysis, use the smart chunked full-document scripts. The other scripts are mainly for alternative chunking settings or quick text-level testing.

7.3 Running the Model

After selecting the model script, click the **Run** button in PyCharm. The console will display the model loading and processing steps. A successful run is indicated by messages such as `Single-label inference complete` and `Process finished with exit code 0`, and the output file will be generated automatically.



```
Run model5_smartchunked_fulldoc
/usr/local/bin/python3.14 /Users/junyi/Downloads/Programming/Python/rdtll/BERT_training/test_model/model5_smartchunked_fulldoc.py
Loading Model-3 (single-label)...
Loading weights: 100%|██████████| 201/201 [00:00<00:00, 5969.33it/s]
Using device: mps
Processing niue_laws_vol2_ul2022.clean.chunks.jsonl
Saved → niue_laws_vol2_ul2022.clean.chunks_model5_singlelabel.csv
Single-label inference complete.
Process finished with exit code 0
```

Figure 39: Example of successful model execution

8 Viewing Model Output Results

After running the selected model script, the prediction results will be saved automatically in the output folder.

1. Locate the output directory

All prediction results are stored in:

`test_model/test_output`

Inside this folder, results are organized by model (Figure 40):

- `model_3`
- `model_4`
- `model_5`

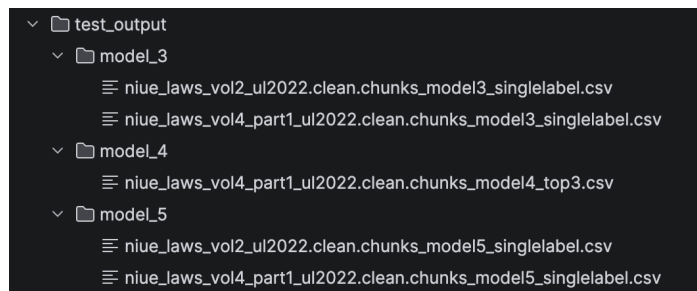


Figure 40: Output folder structure for different models

2. Open a result file

Each model generates output files in CSV format. The file name indicates:

- the document name
- the processing stage (cleaned + chunked)
- the model used

For example:

niu_laws_vol2_ul2022.clean.chunks_model5_singlelabel.csv

3. Single-label output format (Model 3 / Model 5)

For single-label models, each text chunk is assigned one predicted label.

niu_laws_vol2_ul2022.clean.chunks_model5_singlelabel						
doc_id	pdf_pages	section	title	text	predicted_label	confidence
niu_laws_vol2_ul2022	225,226	11	Defences	In any proceedings for an offence under this Act it shall be a sufficient defence for the defendant to prove - (a) That he purchased the food from another person who furnished a written warranty in compliance with section 5, and sold the food in the same condition the article was in at the time he purchased it; and	9_3	0.9784
niu_laws_vol2_ul2022	476	49	Transitional	For the avoidance of doubt, where any of the International Conventions referred to in section 47 of this Act has been purportedly brought into effect by regulations made under this Act, then that Convention shall be deemed to have been given the force of law from that date and all actions taken pursuant to the terms of that Convention shall be deemed to have been validly taken, and any such regulations shall be deemed to have been validly promulgated.	12_10	0.9751
niu_laws_vol2_ul2022	212,213	22	Agreements for information exchange with outside agencies	(1) In this section "agreement" means an agreement or arrangement in writing that relates to the exchange of information between the Unit and any law enforcement agency or supervisory body outside Nue. (2) The Unit may, with the Minister's approval, enter into an agreement with any such body.	6_5	0.9712
niu_laws_vol2_ul2022	27,28	1	Tariff protection	Import duty may be increased by an amount on a class of commodity presently or proposed to be produced or supplied by an enterprise to protect an approved activity where (a) the activity is import substituting; and (b) the protection is essential for the initial survival of the approved activity;	1_4	0.9688
niu_laws_vol2_ul2022	61	11	Prohibited exports	(1) Cabinet may restrict the export of any species of fish and or their meat or body parts by regulation. (2) For the purpose of preventing the export of any species referred to in sub-section (1), an export licence, a restriction order, or an export licence order may be issued.	10_4	0.9669
niu_laws_vol2_ul2022	431	16	When master may be exonerated from liability	PART 3 CAPTAINS 17-20 [Proposed] PART 4 DELIVERY OF GOODS AND LIEN FOR FREIGHT	12_2	0.9612
niu_laws_vol2_ul2022	440	37	Lien of registered transferee of warrant determined on delivery of warrant bona fide and for value	In the event of any transfer being entered in the books of the warehouse keeper, and the then owner of bonded goods delivers over the warrants or certificates relating to or affecting the same to any other person on a sale or pledge of the said goods, and such warrants or certificates are afterwards delivered over bona fide and for value to any subpurchaser or otherwise by the person acquiring the same from the owner whose name is entered on either	12_2	0.961
niu_laws_vol2_ul2022	553	35	Requests for information gathering orders	35A Cabinet may approve proposals for sharing certain property PART 7 REQUESTS BY OR ON BEHALF OF A DEFENDANT	5_1	0.9603
niu_laws_vol2_ul2022	559,560	13	Requests by Nue for search and seizure	(1) This section applies to a proceeding or investigation relating to a criminal matter involving a serious offence against the law of Nue if there are reasonable grounds to believe that a thing relevant to the proceeding or investigation may be located in a foreign country.	7_5	0.9583

Figure 41: Example of single-label prediction output

Each row represents one chunk of text and contains:

- doc_id: document name
- pdf_pages: page reference
- section: detected section
- title: section title
- text: extracted content
- predicted_label: assigned label
- confidence: prediction confidence score

4. Multi-label output format (Model 4)

For multi-label models, each chunk can have multiple predicted labels.

nlw_laws_v04_part1_u02022_clean_chunks_model4_top3										
doc_id	pdf_pages	section	title	text	prediction_1	score_1	prediction_2	score_2	prediction_3	score_3
nlw_laws_v04_part1_u02022	62	92	Accounts, payments and retention of records	(1) The licensee of a ship station open for public correspondence shall keep full accounts, records and registers of all radiotelegrams transmitted by him. Each radiotelegram shall be identified by a number and date. Full particulars of its place of origin and of ultimate destination, and such further particulars as the Superintendent may require to be shown. (2) The licensee shall preserve all such records/accounts from whatever written.	7,3	0.6882	6,2	0.04	5,4	0.017
nlw_laws_v04_part1_u02022	55	52	Log to be kept	(1) The licensee of every transmitting station shall, unless exempted by the Superintendent, keep a log record showing the hours during which the station is in operation, the time of each transmission, the class of emission, the station called, and the power and the frequency used. (2)	7,3	0.6148	6,2	0.0308	8,3	0.003
nlw_laws_v04_part1_u02022	23	3	3 of the Act:	(2) * includes any area, field, farm, garden, orchard, nursery, hothouse, shadehouse, cool store, dwellinghouse, shop, building, room or other place or premises and references to land extend to and include any harbour, highway, road, wharf, port or airport. quarantine control.	10,4	0.6014	10,1	0.0851	10,2	0.064
nlw_laws_v04_part1_u02022	85,98	12	Directors for preparation of financial statements	Financial statements of a company prepared for the purposes of the Act must comply with the following accounting policies: Accrual accounting	5,4	0.5292	7,3	0.0777	11,2	0.026
nlw_laws_v04_part1_u02022	54	46	Allocation of callign	The Superintendent shall allot to every transmitting station licensed under these Regulations a callign by which the station shall be identified.	8,3	0.5146	9,4	0.0681	12,3	0.040
nlw_laws_v04_part1_u02022	35	13	Powers of Corporation to conduct inquiries	(1) The Corporation shall have the power to conduct inquiries as may be necessary to ensure compliance with these Regulations. (2)	7,5	0.466	9,1	0.082	8,4	0.080
nlw_laws_v04_part1_u02022	89,90	79	Completed by	Without limitation the powers of the Corporation under subsection (1) be: Address Email Telephone Facsimile Fax Optional	3,3	0.466	3,5	0.0543	12,8	0.036
nlw_laws_v04_part1_u02022	33	3	Sitting allowances for Directors	(1) Every Director of the Corporation shall be paid an allowance of \$40 per meeting for every meeting that Director attends. (2) The allowances payable to the Directors shall be a charge upon the funds of the Corporation.	3,3	0.4395	3,5	0.0765	12,8	0.058

Figure 42: Example of multi-label prediction output (Top 3 labels)

In this case, the output includes:

- prediction_1, score_1
- prediction_2, score_2
- prediction_3, score_3

This allows the model to provide multiple possible classifications ranked by confidence.

5. How to interpret the results

- A higher confidence score indicates stronger model certainty.
- Single-label outputs are simpler and easier to use for structured analysis.
- Multi-label outputs are useful when legal text may belong to multiple categories.

For a better understanding, it is recommended to read and check RDTII_Legal_NLP_Project_Documentation, section 7.5 Confidence Score Interpretation.

6. Recommended usage

- Use **Model 5 output** for standard single-label analysis.
- Use **Model 4 output** when multi-label interpretation is needed.
- Compare Model 3 and Model 5 if further validation or diagnostic analysis is required.

In practice, the CSV output files **cannot** be opened correctly in **Microsoft Excel** using default settings.

Therefore, it is recommended to check the results in **PyCharm** or, for Mac users, in **Numbers**. For a more detailed explanation of this issue, see RDTII_Legal_NLP_Project_Documentation, section 7.4 Excel Compatibility.